

Customer, Product, and Profitability Performance Analysis in Supply Chain Operations

Project Documentation & Business Analysis

1. Project Objective

Analyze customer, product, sales, profitability, shipping, regional and delivery performance using data preparation, exploratory analysis and an interactive Power BI dashboard.

2. Tools & Technologies

- Python / Jupyter Notebook — data inspection, cleaning, preparation and exploratory analysis.
- Power BI — KPI creation, interactive visualization, filtering and dashboard development.
- Cleaned supply-chain transactional dataset.

3. Data Preparation

The data was inspected, cleaned and prepared for visualization. The final analytical fields include Sales, Order Profit Per Order, Customer Id, Category Name, Order Region, Shipping Mode, Delivery Status and Shipping_Delay.

4. Dashboard KPIs

KPI	Value	Purpose
Total Sales	36.78M	Overall sales performance
Total Profit	3.97M	Overall profitability
Total Customers	20.652K	Distinct customer base
Average Shipping Delay	0.57	Delivery/logistics performance

5. Power BI Visualizations

- Sales by Category
- Sales by Region
- Sales by Shipping Mode
- Sales by Delivery Status
- Shipping Delay by Delivery Status
- Average Profit per Order by Shipping Mode
- Sales vs Profit by Product
- Order Region slicer for interactive filtering

6. Key Business Insights

- Total Sales are 36.78M and Total Profit is 3.97M.
- The dashboard reports 20.652K distinct customers.
- Fishing is the leading sales category in the dashboard.
- Western Europe is the strongest region by sales among the displayed regions.
- Standard Class generates the largest sales volume, approximately 22M.
- Late Delivery has the largest sales contribution among the displayed delivery statuses, making delivery performance an important operational focus.
- Average Shipping Delay is 0.57.
- First Class shows the highest average profit per order among the displayed shipping modes.

7. Business Recommendations

- Reduce late deliveries through better warehouse processing, transportation planning and shipment monitoring.
- Study high-performing regional practices and apply relevant strategies to lower-performing regions.
- Maintain Standard Class volume while evaluating the profitability of faster shipping modes.
- Review low-performing categories for pricing, demand, inventory and promotional opportunities.
- Monitor profit as well as sales volume when making supply-chain decisions.
- Use the region slicer to investigate regional differences.

8. Conclusion

The project combines data preparation, exploratory analysis and Power BI visualization to provide a consolidated view of supply-chain performance. The results show strong sales and positive profitability, while delivery performance—particularly late delivery—presents a key opportunity for operational improvement.

9. Project Status

Data preparation: Completed | Power BI dashboard: Completed | KPI section: Completed | Business insights: Documented